

5.5 Classwork: Cumulative Review

1. An arithmetic sequence has first term $u_1 = 5$ and common difference $d = 3$.

- (a) Find u_{20} , the 20th term of the sequence. [2]

- (b) Find S_{20} , the sum of the first 20 terms. [2]

A geometric sequence has the same first term $u_1 = 5$ and common ratio $r = 1.1$.

- (c) Find the 20th term of the geometric sequence. [2]

- (d) Find the sum of the first 20 terms of the geometric sequence. [2]

- (e) After how many terms does the geometric sequence first exceed 50? Show your working. [3]

x (hours)	1.5	2.0	3.0	3.5	4.0	5.0	5.5	6.0
y (score)	42	48	56	55	65	68	74	78

- Use your GDC to find the equation of the regression line y on x . Write the equation in the form $y = mx + c$, giving m and c to three significant figures. [3]
- Write down the value of the Pearson correlation coefficient, r . [1]
- Describe the correlation between hours studied and test score. [1]
- Use your regression line to estimate the test score for a student who studies for 4.5 hours. [2]
- Explain why it would not be appropriate to use this model to predict the score for a student who studies for 15 hours. [1]

3. A particle moves along a straight line with velocity $v(t) = 3t^2 - 12t + 9 \text{ m s}^{-1}$, for $t \geq 0$.
- (a) Find the times when the particle is at rest. [3]
- (b) Find the average acceleration from $t = 0$ to $t = 1$. [2]
- (c) Find the velocity when $t = 2$. State which direction the particle is moving at this instant. Justify your answer. [3]

4. In a class of 30 students, 18 study Biology and 15 study Chemistry. Every student studies at least one of these subjects.

(a) Find the number of students who study both Biology and Chemistry. [2]

(b) Draw a Venn diagram to represent this information. [2]

A student is chosen at random from the class. Let B be the event “the student studies Biology” and C be the event “the student studies Chemistry.”

(c) Find $P(B \cap C)$. [1]

(d) Find $P(B \mid C)$. [2]

(e) Determine whether the events B and C are independent. Justify your answer. [3]

It is known that 70% of the Biology students and 40% of the Chemistry-only students passed their most recent exam.

(f) A student is chosen at random and is found to have passed. Find the probability that this student studies Biology. [4]